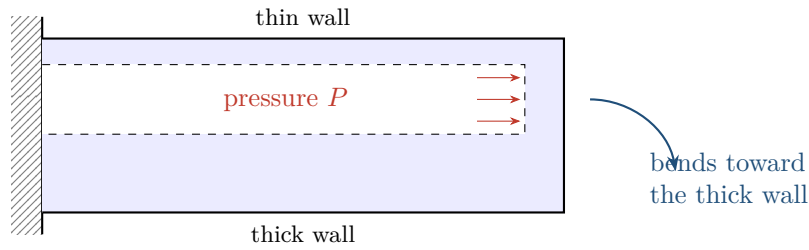


B.Tech Project

Pressure-to-Deflection Modelling of an Eccentric-Cavity Soft Pneumatic Bending Actuator

Analytical derivation and verification
by nonlinear finite-element analysis



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Abstract. A silicone cylinder with an off-centre (eccentric) cavity bends when the cavity is pressurised. This document derives the pressure-to-deflection relation of the actuator. The cross-section properties, the eccentric thrust moment and the Euler–Bernoulli response are derived step by step, and the validity of each result is stated. Two physical results, both confirmed by a verified 3-D nonlinear finite-element (FE) model, decide how the beam result is used. First, for a bare incompressible actuator the beam formula is multiplied by $(1 - 2\nu)$, which is zero for silicone. Second, the bending a bare actuator does show is a finite-strain ballooning effect that is quadratic in P at low pressure and ends at a limit pressure. The beam formula is therefore the correct first-order model for a *hoop-reinforced* actuator.

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Units Units throughout: mm, N, MPa ($1 \text{ MPa} = 1000 \text{ kPa} = 1 \text{ N/mm}^2$). Boxed equations are the main results; blue boxes are remarks.

1 Introduction

1.1 The device

The actuator is a solid silicone cylinder of outer radius R and pressurised length l . A circular cavity of radius r runs along it, with its centre offset by r_p from the axis of the cylinder. One wall is therefore thin, $t_{\min} = R - r_p - r$, and the opposite wall is thick, $t_{\max} = R + r_p - r$. The base is clamped, the far end is sealed by a solid cap, and the cavity is pressurised to a gauge pressure P . The actuator curls toward the thick wall (Figure 1).

Objective. Find the map from the input P to the tip bending angle θ and the tip deflection δ .

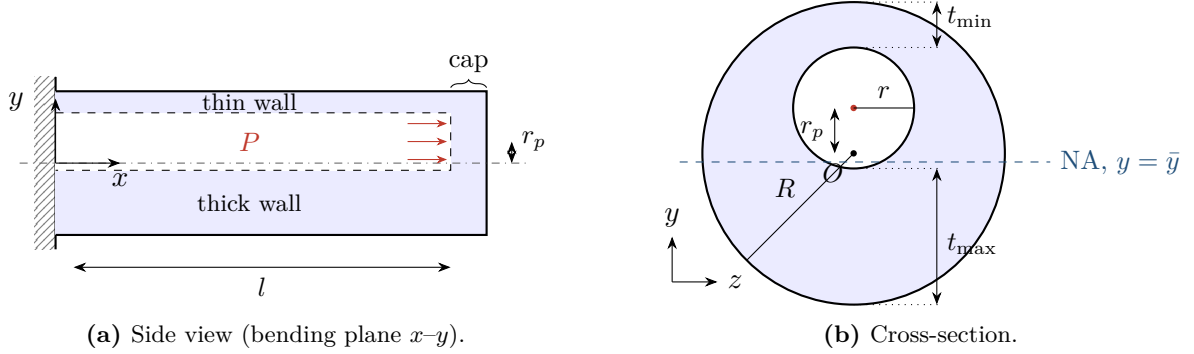


Figure 1: Geometry and coordinate frame. The origin O is the centre of the outer circle, the cavity centre is at $(y, z) = (r_p, 0)$ and the neutral axis (NA) lies at $y = \bar{y} < 0$. Not to scale.

1.2 Scope

Sections 3–5 derive the cross-section properties, the pressure loading and the beam response. Section 6 adds the effect of the in-plane wall stresses, which decides whether a bare actuator bends at first order. Section 7 analyses the asymmetric expansion of the thin and thick walls. Section 8 lists the model hierarchy and Section 9 summarises the FE verification. All numerical values refer to the *placeholder* design $R = 10$, $r = 4$, $r_p = 3$, $l = 100$ mm with a neo-Hookean silicone of $E = 0.6$ MPa.

2 Notation and assumptions

Symbol	Meaning	Placeholder value
R	outer radius	10 mm
r	cavity radius	4 mm
r_p	cavity eccentricity (offset of the cavity centre along $+y$)	3 mm
l	pressurised length	100 mm
$t_{\min}, t_{\max}$	thinnest and thickest wall, $R \mp r_p - r$	3 mm, 9 mm
E, μ, ν	small-strain Young's modulus, shear modulus, Poisson's ratio	0.6 MPa, 0.2 MPa, ≈ 0.5
C_{10}, C_{20}, C_{30}	Yeoh coefficients (neo-Hookean: $C_{10} = \mu/2 = E/6$ only)	$C_{10} = 0.1$ MPa
P	gauge pressure in the cavity (the input)	0 kPa to 110 kPa
A, A_c	elastomer area, cavity area πr^2	
$\bar{y}, I_{zz}, e$	neutral-axis position, second moment about it, moment arm	
κ, θ, δ	curvature, tip angle, tip deflection	

Table 1: Symbols. Axes: x along the actuator (base at $x = 0$, tip at $x = l$); y in the section, from the outer centre toward the cavity centre; z completes the right-handed frame. Bending is about z , toward $-y$.

Assumptions.

- (A1) The material is isotropic, homogeneous, rate-independent and (nearly) incompressible; Mullins and viscoelastic effects are removed by preconditioning.
- (A2) The pressure is uniform in the cavity; gravity and external loads are neglected; the base is clamped and the tip is free.
- (A3) *Beam model only*: plane sections remain plane, the cross-section keeps its shape, and every longitudinal fibre is in uniaxial stress. Assumption (A3) holds for a hoop-reinforced actuator and fails for a bare one (Section 6).

3 Cross-section geometry

3.1 Elastomer area

The section is the outer disc minus the cavity disc:

$$A = \pi R^2 - \pi r^2 = \pi (R^2 - r^2). \quad (3.1)$$

3.2 Neutral axis

Treat the cavity as a negative area centred at $y = r_p$. The centroid of the composite area is

$$\bar{y} = \frac{\pi R^2 \cdot 0 - \pi r^2 \cdot r_p}{\pi R^2 - \pi r^2} \implies \boxed{\bar{y} = -\frac{r^2 r_p}{R^2 - r^2}} \quad (3.2)$$

The neutral axis sits on the thick-wall side of the outer centre ($\bar{y} < 0$), because material has been removed on the $+y$ side. Dimensional check: $r^2 r_p / (R^2 - r^2)$ is a length.

3.3 Second moment of area about the neutral axis

By the parallel-axis theorem $I = I_c + Ad^2$ applied to each disc, with distances measured from the neutral axis:

$$I_{zz} = \left[\frac{\pi R^4}{4} + \pi R^2 \bar{y}^2 \right] - \left[\frac{\pi r^4}{4} + \pi r^2 (r_p - \bar{y})^2 \right]. \quad (3.3)$$

To simplify, note that $r_p - \bar{y} = r_p + \frac{r^2 r_p}{R^2 - r^2} = \frac{R^2 r_p}{R^2 - r^2}$. Then

$$\pi R^2 \bar{y}^2 - \pi r^2 (r_p - \bar{y})^2 = \frac{\pi r_p^2}{(R^2 - r^2)^2} \left[R^2 r^4 - r^2 R^4 \right] = -\frac{\pi R^2 r^2 r_p^2 (R^2 - r^2)}{(R^2 - r^2)^2} = -\frac{\pi R^2 r^2 r_p^2}{R^2 - r^2}, \quad (3.4)$$

so

$$\boxed{I_{zz} = \frac{\pi (R^4 - r^4)}{4} - \frac{\pi R^2 r^2 r_p^2}{R^2 - r^2}} \quad (3.5)$$

The first term is the concentric hollow cylinder; the second is the stiffness lost by shifting the hole off-centre. For the placeholder design $I_{zz} = \pi(2436 - 171.43) = 7114.4 \text{ mm}^4$ (the parallel-axis form evaluated numerically gives the same value).

4 Pressure loading

4.1 Thrust on the sealed tip

The pressure acts on the inside face of the tip cap, whose projected area is the cavity area $A_c = \pi r^2$:

$$F = P A_c = P \pi r^2. \quad (4.1)$$

Note that the thrust uses the cavity area A_c , not the elastomer area A of eq. (3.1).

4.2 Moment arm

The resultant of a uniform pressure on a circular face passes through the centre of the circle, $y = r_p$. Its distance from the neutral axis is

$$e = r_p - \bar{y} = r_p + \frac{r^2 r_p}{R^2 - r^2} \implies \boxed{e = \frac{R^2 r_p}{R^2 - r^2}} \quad (4.2)$$

4.3 Bending moment

$$\boxed{M = F e = \frac{\pi P R^2 r^2 r_p}{R^2 - r^2}} \quad (4.3)$$

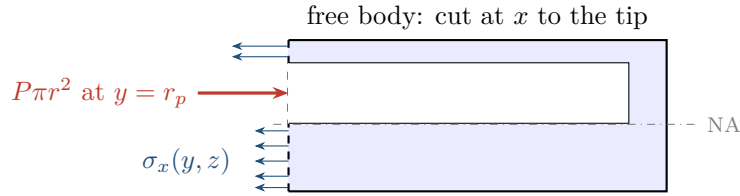


Figure 2: Free body from a cut at x to the tip, silicone together with the fluid it encloses. The fluid beyond the cut pushes with $P\pi r^2$ through the cavity centre; the silicone carries the axial stress σ_x .

Why M is uniform along the length Cut the actuator at any x and take the free body from the cut to the tip, *including the fluid inside it* (Figure 2). No external force acts on it except at the cut. Equilibrium gives

$$\int_A \sigma_x dA = P\pi r^2, \quad \int_A \sigma_x (y - r_p) dA = 0. \quad (4.4)$$

So the wall is in net tension $N = P\pi r^2$ and the stress resultant acts through the cavity centre. About the neutral axis this is the moment $M = Ne$ of eq. (4.3), at every x and for any material law. Eq. (4.4) fixes the *resultants* of σ_x only; turning them into strain needs a constitutive assumption (Sections 5 and 6).

5 Beam-theory response (Euler–Bernoulli)

Under assumption (A3), $\sigma_x = E \varepsilon_x$ with $\varepsilon_x = \varepsilon_0 + \kappa (y - \bar{y})$.

5.1 Curvature

$$\boxed{\kappa = \frac{1}{\rho} = \frac{M}{EI_{zz}} = \frac{\pi P R^2 r^2 r_p}{E I_{zz} (R^2 - r^2)}} \quad (5.1)$$

Here E is the small-strain modulus, $E = 3\mu = 6C_{10}$ for an incompressible solid. A secant modulus chosen to fit a curve would hide the real nonlinearity, which for the bare actuator is ballooning (Section 7), not a softening of E .

5.2 Tip bending angle

M is uniform, so κ is uniform and the centreline is a circular arc:

$$\theta = \kappa l = \frac{Ml}{EI_{zz}} = \frac{\pi P R^2 r_p l}{E I_{zz} (R^2 - r^2)} \quad (5.2)$$

The tip angle is linear in P , since κ is.

5.3 Tip deflection

For an arc of radius $\rho = 1/\kappa$ turning through θ , the tip moves sideways by

$$\delta = \rho (1 - \cos \theta) = \frac{1 - \cos \theta}{\kappa} = \frac{l}{\theta} (1 - \cos \theta). \quad (5.3)$$

For $\theta \ll 1$, $1 - \cos \theta \approx \theta^2/2$ and

$$\delta \approx \frac{\theta l}{2} = \frac{M l^2}{2 E I_{zz}}, \quad (5.4)$$

which is the cantilever result for a uniform end moment. The axial tip position is $x_{\text{tip}} = \rho \sin \theta$.

5.4 Axial elongation

The net tension of eq. (4.4) stretches the centroidal fibre by $\varepsilon_0 = F/(EA)$, so $\Delta l = Fl/(EA)$. A uniform axial strain moves every fibre by the same amount and does not shift the neutral axis. This elongation holds under assumption (A3), i.e. for a hoop-reinforced actuator; for a bare incompressible actuator the first-order elongation is $(1 - 2\nu)Fl/(EA) \approx 0$ (Section 6).

5.5 Non-dimensional form

Writing $\hat{I} = I_{zz}/R^4$,

$$\theta = \frac{P}{E} \frac{l}{R} f\left(\frac{r}{R}, \frac{r_p}{R}\right), \quad f = \frac{\pi (r/R)^2 (r_p/R)}{\hat{I} (1 - (r/R)^2)}. \quad (5.5)$$

The response is linear in P/E and in the slenderness l/R . The section shape enters only through f . For the placeholder design $\hat{I} = 0.7114$ and $f = 0.2523$.

5.6 Worked example

Section (eqs. 3.1–4.2)		Response at $P = 20$ kPa (eqs. 4.1–5.4)	
A	263.9 mm ²	F	1.005 N
$\bar{y}$	−0.571 mm	M	3.590 N mm
I_{zz}	7114.4 mm ⁴	κ	8.41×10^{-4} mm ^{−1}
e	3.571 mm	θ	4.82°
$t_{\min}, t_{\max}$	3 mm, 9 mm	δ arc / small-angle	4.203 mm / 4.206 mm

Table 2: Beam model for the placeholder design, $E = 0.6$ MPa. At 10 kPa, 40 kPa and 60 kPa: $\theta = 2.41^\circ$, 9.64° and 14.46° , $\delta_{\text{arc}} = 2.10$ mm, 8.39 mm and 12.55 mm.

Validity of eqs. (5.1)–(5.4). They rest on assumption (A3): each fibre carries only axial stress. That is true when a hoop-fibre sleeve (thread wrap) carries the hoop load. The FE model confirms that for the *sleeved* actuator the tip angle matches eq. (5.2) within 5 % up to $P/E \approx 0.03$. For the *bare* actuator see the next section.

6 The bare actuator: effect of the in-plane wall stresses

The pressure does not only push on the tip cap. It also loads the wall *in the plane of the section*: radial compression at the cavity surface and hoop tension around it, both of order P or larger. Assumption (A3) ignores them. In linear isotropic elasticity they enter the axial strain through Poisson's ratio:

$$\varepsilon_x = \frac{\sigma_x - \nu(\sigma_y + \sigma_z)}{E} \implies \sigma_x = E[\varepsilon_0 + \kappa(y - \bar{y})] + \nu(\sigma_y + \sigma_z). \quad (6.1)$$

6.1 Resultants of the in-plane stresses

The in-plane field need not be solved: its resultant and first moment follow from equilibrium alone. Far from the ends the in-plane stresses $\sigma_{\alpha\beta}$ ($\alpha, \beta \in \{y, z\}$) satisfy $\partial_\beta \sigma_{\alpha\beta} = 0$, the outer surface is traction-free and the cavity surface carries the traction $P\mathbf{m}$, where $\mathbf{m}$ is the unit normal pointing out of the cavity into the wall. Take coordinates $\mathbf{x}'$ centred on the cavity. From the identities

$$\partial_\beta(\sigma_{\alpha\beta}x'_\alpha) = \sigma_{\alpha\alpha}, \quad \partial_\beta(\sigma_{\alpha\beta}x'_\alpha x'_\gamma) - \partial_\beta(\sigma_{\gamma\beta} \tfrac{1}{2}|\mathbf{x}'|^2) = \sigma_{\alpha\alpha} x'_\gamma$$

and the divergence theorem, with $\mathbf{x}' = r\mathbf{m}$ on the cavity boundary,

$$\int_A (\sigma_y + \sigma_z) dA = \oint P r ds = 2P\pi r^2, \quad \int_A (\sigma_y + \sigma_z) x'_\gamma dA = \oint P r^2 m_\gamma ds - \oint \frac{Pr^2}{2} m_\gamma ds = 0. \quad (6.2)$$

So the in-plane stresses have a total of $2P\pi r^2$ acting *through the cavity centre*, for an outer boundary of any shape. About the neutral axis:

$$\int_A (\sigma_y + \sigma_z) dA = 2P\pi r^2, \quad \int_A (\sigma_y + \sigma_z)(y - \bar{y}) dA = 2P\pi r^2 e. \quad (6.3)$$

6.2 Corrected first-order response

Substitute eq. (6.1) into the two free-body conditions (4.4), using $\int_A (y - \bar{y}) dA = 0$:

$$\text{force: } EA\varepsilon_0 + 2\nu P\pi r^2 = P\pi r^2, \quad (6.4)$$

$$\text{moment: } EI_{zz}\kappa + 2\nu P\pi r^2 e = P\pi r^2 e. \quad (6.5)$$

$$\varepsilon_0 = (1 - 2\nu) \frac{P\pi r^2}{EA}, \quad \kappa = (1 - 2\nu) \frac{M}{EI_{zz}} = (1 - 2\nu) \frac{\pi P R^2 r^2 r_p}{E I_{zz} (R^2 - r^2)}. \quad (6.6)$$

For silicone $\nu \rightarrow 0.5$: **in the small-strain limit a bare eccentric-cavity actuator neither lengthens nor bends.**

The first result is the textbook statement that a closed incompressible tube does not lengthen under internal pressure; the second is its bending counterpart. The thrust moment is still there, but the Poisson effect of the in-plane stresses cancels it exactly. The beam results of Section 5 therefore apply to the hoop-reinforced actuator, and eq. (6.6) to the bare one.

FE check of eq. (6.6) Mid-span curvature from the FE model divided by eq. (5.1), at $P/E = 10^{-5}$:

ν	0	0.2	0.3	0.4	0.45	0.4995
FE	1.0000	0.6001	0.4001	0.2002	0.1002	0.0012
$1 - 2\nu$	1.0000	0.6000	0.4000	0.2000	0.1000	0.0010

The tip angle of the nearly incompressible actuator is not exactly zero but 6 % of eq. (5.2). All of it comes from the two ends, where the clamp and the cap stop the wall from expanding and switch the cancellation off locally.

7 Asymmetric wall expansion

There is a second bending mechanism. The thin wall stretches more in hoop than the thick wall, the stretch couples into axial lengthening, and the thin side becomes longer than the thick side. This section derives the wall geometry, shows why a linear estimate of this effect fails, and gives the correct finite-strain result.

7.1 Wall thickness around the cavity

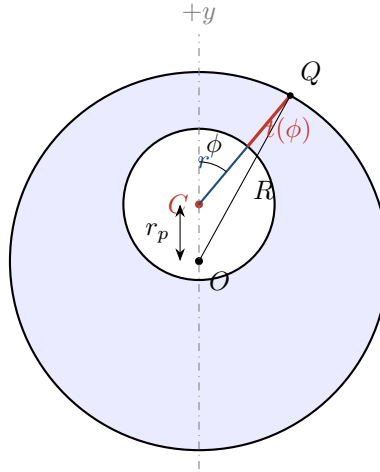


Figure 3: Wall thickness along a ray from the cavity centre C at angle ϕ from $+y$. $\phi = 0$ points at the thin wall.

Along a ray from the cavity centre C at angle ϕ from $+y$ (Figure 3), the point Q at distance $s = r + t$ lies on the outer circle. The law of cosines in triangle OCQ gives

$$R^2 = r_p^2 + s^2 + 2r_p s \cos \phi \implies s = -r_p \cos \phi + \sqrt{R^2 - r_p^2 \sin^2 \phi}, \quad (7.1)$$

taking the positive root, so

$$t(\phi) = \sqrt{R^2 - r_p^2 \sin^2 \phi} - r_p \cos \phi - r \quad (7.2)$$

with $t(0) = R - r_p - r = t_{\min}$ and $t(\pi) = R + r_p - r = t_{\max}$. The expression is exact for the radial wall thickness along the ray. The FE mesh is built on this formula. As before, the longitudinal strain and stress are ε_x and σ_x , since the actuator axis is x .

7.2 Hoop stress

In the thin-membrane approximation the hoop stress at angle ϕ is $\sigma_\theta(\phi) \approx Pr/t(\phi)$, and the mean axial stress is $\bar{\sigma}_x = P\pi r^2/A$. The membrane estimate needs $t \ll r$; here $t_{\min}/r = 0.75$, so the values are indicative only.

7.3 A linear estimate, and why it fails

A first guess takes $\varepsilon_x(\phi) \propto Pr/(Et(\phi))$ and defines

$$\kappa_{\text{exp}} \approx \frac{\varepsilon_x(\text{thin}) - \varepsilon_x(\text{thick})}{2R} = \frac{Pr}{2ER} \left(\frac{1}{R - r_p - r} - \frac{1}{R + r_p - r} \right), \quad (7.3)$$

to be added to the thrust curvature, $\kappa_{\text{total}} = \kappa_{\text{thrust}} + \kappa_{\text{exp}}$. This estimate does not hold:

1. It rests on linear Poisson coupling, and Section 6 shows that for $\nu = 0.5$ linear elasticity gives *no* axial strain and *no* curvature from pressure at all. The linear in-plane effect is exactly what cancels κ_{thrust} ; it cannot also add to it.
2. The superposition $\kappa_{\text{total}} = \kappa_{\text{thrust}} + \kappa_{\text{exp}}$ is linear in P and equals $2.76 \kappa_{\text{thrust}}$ at every pressure for the placeholder design. The FE curvature of the bare actuator runs from 0.1 to 3 times κ_{thrust} over the pressure range: it is not linear in P .

7.4 The correct mechanism: second-order lengthening

Consider a thin closed neo-Hookean tube with hoop, axial and thickness stretches $\lambda_1 = 1 + a$, $\lambda_2 = 1 + b$, $\lambda_3 = 1/(\lambda_1\lambda_2)$. The membrane stresses are $\sigma_\theta = \mu(\lambda_1^2 - \lambda_3^2)$ and $\sigma_x = \mu(\lambda_2^2 - \lambda_3^2)$, and a closed end gives $\sigma_x = \sigma_\theta/2$:

$$2\lambda_2^2 = \lambda_1^2 + \lambda_3^2. \quad (7.4)$$

Expanding with $\lambda_3^2 \approx 1 - 2a - 2b + 3a^2 + 3b^2 + 4ab$:

- first order: $4b = -2b$, so $b = 0$. There is no axial strain at first order, in agreement with eq. (6.6);
- second order, with $b = O(a^2)$: $6b = 4a^2$, so

$$\varepsilon_{\text{axial}} \approx \frac{2}{3} \varepsilon_{\text{hoop}}^2 \quad (7.5)$$

The thin wall has the larger hoop strain, so it lengthens more and the actuator bends toward the thick wall. The wall-expansion mechanism is therefore real, but it is *quadratic* in P , not linear.

Scaling estimate Take the closed-tube membrane hoop strain $\varepsilon_{\text{hoop}} = \frac{3}{4} Pr/(Et)$ for each wall, apply eq. (7.5), and divide the difference in axial strain by the distance between the wall mid-lines ($d = 14 \text{ mm}$):

$$\kappa_{\text{wall}} \approx \frac{2}{3} \frac{\varepsilon_{\text{hoop}}(t_{\min})^2 - \varepsilon_{\text{hoop}}(t_{\max})^2}{d}.$$

This gives $1.2 \times 10^{-5} \text{ mm}^{-1}$ at 10 kPa and $4.7 \times 10^{-5} \text{ mm}^{-1}$ at 20 kPa, against FE mid-span values of 1.8×10^{-5} and $7.8 \times 10^{-5} \text{ mm}^{-1}$. The estimate is within a factor of two and reproduces the P^2 scaling: the FE curvature grows by 4.3 when P doubles. It is a rough guide only, since the wall is not thin and the two walls interact.

7.5 Ballooning limit

A rubber tube under internal pressure has a pressure maximum: past it, the wall thins faster than the hoop stress can rise, and the tube inflates without bound at falling pressure. The thin wall

reaches it first. For the placeholder shape the FE model gives the limit at

$$P_{\text{lim}} = 0.523 \mu \quad (104.7 \text{ kPa for } E = 0.6 \text{ MPa; } 14 \text{ kPa for } E = 0.08 \text{ MPa}). \quad (7.6)$$

Above P_{lim} a neo-Hookean actuator has no equilibrium under pressure control. Under volume control (a water-filled syringe) the path through the limit is stable. A real silicone stiffens at large stretch (Yeoh $C_{20}, C_{30} > 0$), which can raise or remove the limit, so the material must be measured before this number is trusted. In short, wall compliance dominates the bare actuator: it is the only bending mechanism that survives, and it ends in a limit pressure.

8 Model hierarchy

Tier	Content	Applies to
1	Linear Euler–Bernoulli, eqs. (5.1)–(5.4)	hoop-reinforced, $P/E \lesssim 0.03$
2	Tier 1 curvature with large-rotation arc kinematics and axial stretch: $l_c = l(1 + \varepsilon_0)$, $\rho = l_c/\theta$, $x_{\text{tip}} = \rho \sin \theta$, $\delta = \rho(1 - \cos \theta)$	hoop-reinforced; the form needed for control
3	Hyperelastic section equilibrium: $\lambda(y) = \lambda_c + k(y - r_p)$; solve $\int s(\lambda) dA = P\pi r^2$, $\int s(\lambda)(y - r_p) dA = 0$ with $s(\lambda) = 2(\lambda - \lambda^{-2})[C_{10} + 2C_{20}(I_1 - 3) + 3C_{30}(I_1 - 3)^2]$, $I_1 = \lambda^2 + 2/\lambda$; then $\theta = kl$	hoop-reinforced, larger P
4	Nonlinear 3-D finite elements (Section 9)	bare and reinforced, up to the limit point
5	Experiment	material constants and validation

Table 3: Model tiers. Tiers 1–3 share assumption (A3) and describe the hoop-reinforced actuator; the bare actuator needs Tier 4.

9 Verification by nonlinear finite-element analysis

9.1 What was done and how

No commercial structural FE package was available, so a solver was written in Python with NumPy, SciPy and JAX (`actuator_fea.py`):

- **Mesh:** 27-node hexahedra, half model with symmetry about the bending plane (Figure 4); 588 elements (medium mesh); clamped base, solid tip cap of thickness $R/2$.
- **Material:** Yeoh/neo-Hookean with an isochoric–volumetric split, mixed displacement–pressure (Q2/P1) formulation to avoid volumetric locking, $\nu = 0.4995$.
- **Load:** follower pressure on every wetted cavity face, derived from the potential $-PV$ of the enclosed cavity volume.
- **Solver:** Newton iteration with tangents from automatic differentiation; pressure control, switching automatically to volume control when a pressure step has no solution, so the limit point is passed and located.
- **Variants:** bare actuator, and the same actuator with an ideal inextensible hoop-fibre sleeve on the outer surface.

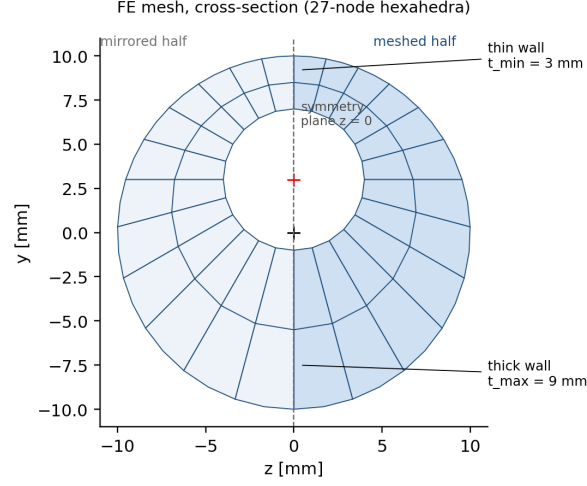


Figure 4: FE mesh of the cross-section. Only the half $z \geq 0$ is solved; the other half follows by symmetry.

9.2 Verification of the solver

Test	Reference	Largest deviation
Uniaxial tension, $\lambda \leq 3$	exact incompressible stress	0.2 % neo-Hookean; 2.5 % Yeoh at $\lambda = 3$
Thick-walled tube inflation, $P/\mu \leq 0.7$	exact neo-Hookean solution	0.05 %
Small-pressure bending, $0 \leq \nu \leq 0.4995$	$\kappa = (1 - 2\nu)M/(EI_{zz})$	0.0002 (in units of eq. 5.1)
Mesh refinement, δ at $P/\mu = 0.30, 0.45$	fine mesh, 1664 elements	medium mesh within 0.5 %

Table 4: Verification (fea_verify.py, log in results/verification.log).

9.3 Findings

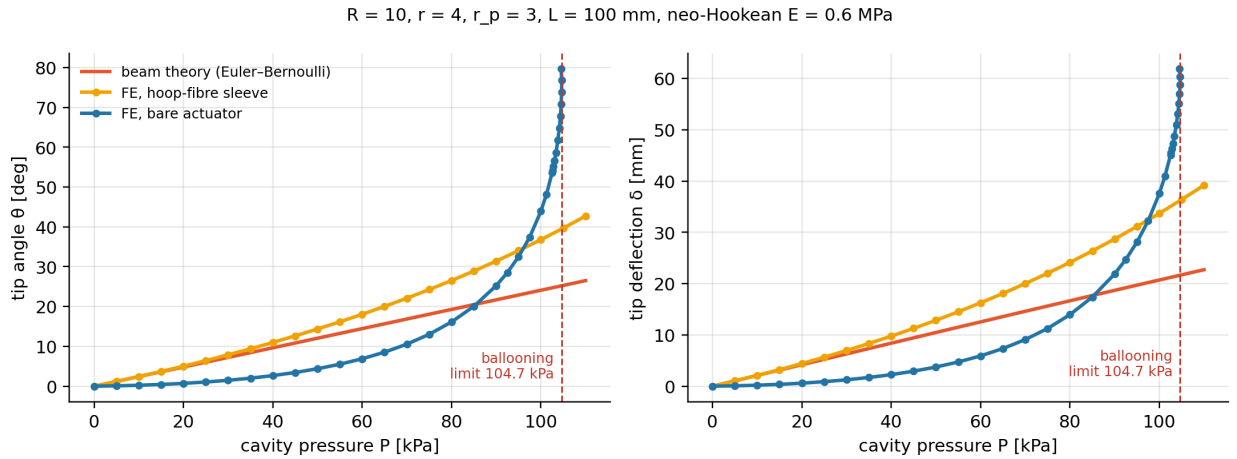


Figure 5: Tip angle and tip deflection against pressure: beam theory (eq. 5.2, 5.3) and FE for the bare and the sleeved actuator.

P [kPa]	θ_{beam} [°]	Bare				Hoop-reinforced			
		θ [°]	FE/beam	δ [mm]	V/V_0	θ [°]	FE/beam	δ [mm]	V/V_0
10	2.41	0.24	0.10	0.20	1.07	2.41	1.00	2.10	1.04
20	4.82	0.71	0.15	0.59	1.14	5.03	1.04	4.42	1.09
40	9.64	2.66	0.28	2.26	1.33	11.0	1.14	9.77	1.19
60	14.5	6.89	0.48	5.91	1.59	18.1	1.25	16.3	1.30
80	19.3	16.1	0.84	14.0	2.01	26.5	1.38	24.2	1.44
100	24.1	44.0	1.82	37.6	2.98	36.8	1.53	33.7	1.61
104.65	25.2	77	3.0	60	4.05	—	—	—	—

Table 5: FE results, $E = 0.6$ MPa ($\mu = 200$ kPa). The last bare row is the ballooning limit. For another neo-Hookean silicone, rescale the pressure by μ .

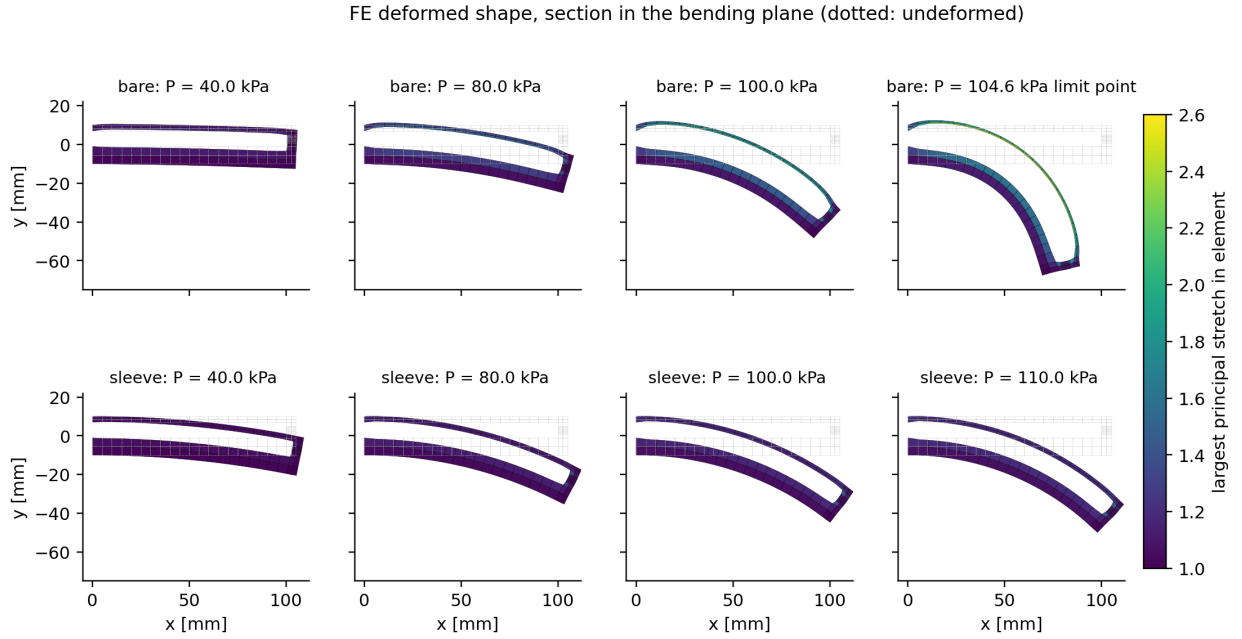


Figure 6: Deformed section in the bending plane, coloured by the largest principal stretch. Top: bare actuator, with the thin wall ballooning near the limit. Bottom: sleeved actuator.

1. **Bare actuator.** Below $P/\mu \approx 0.2$ the beam model over-predicts by a factor of 4 to 10. The curves cross near $P/\mu = 0.43$; that is a coincidence of two different mechanisms, not a validation. The response then steepens without bound: between 100 kPa and the limit, 4.6 kPa adds 23 mm of deflection. At the limit the cavity holds four times its original volume and the thin wall is at a third of its thickness.
2. **Hoop-reinforced actuator.** The response is smooth and nearly linear, with no limit point up to 110 kPa. Eq. (5.2) holds within 5 % up to $P/E \approx 0.03$. At higher pressure the FE angle is larger (by 25 % at $P/E = 0.1$) because the cavity grows as the actuator lengthens inside a sleeve of fixed circumference, and the thrust grows with it.
3. **Design consequence.** Reinforce the actuator if a predictable, nearly linear $P \rightarrow \delta$ map is wanted (e.g. for control). Keep it bare only to study ballooning, and then operate below about 90 % of the measured limit pressure.

10 Conclusions and next steps

1. The cross-section properties (eqs. 3.1–3.5) and the eccentric thrust moment (eqs. 4.1–4.3) follow from statics alone and hold for any material law.

2. The Euler–Bernoulli result, eq. (5.2), is the correct first-order model for a *hoop-reinforced* actuator and has been confirmed by FE.
3. For a *bare* silicone actuator the first-order bending is zero, by the factor $(1 - 2\nu)$. What bending remains comes from second-order wall lengthening, $\varepsilon_{\text{axial}} \approx \frac{2}{3}\varepsilon_{\text{hoop}}^2$, and ends at a ballooning limit $P_{\text{lim}} \approx 0.52 \mu$.
4. Next: fix the real geometry and silicone, measure the Yeoh constants (ASTM D412 dumbbells), choose bare or reinforced, re-run the FE sweep with the measured material, then fabricate and test against it.

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